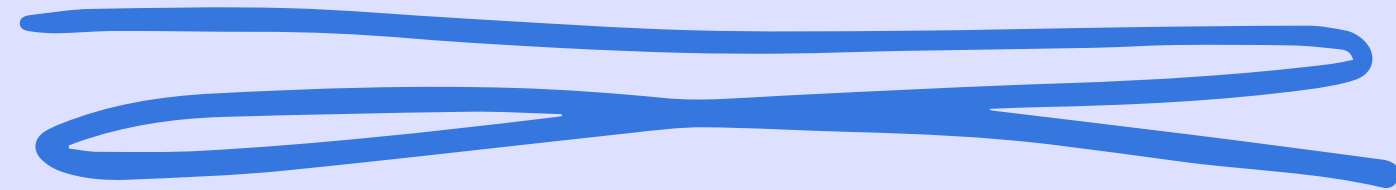


COMPUTER VISION PROJECT 2025/2026

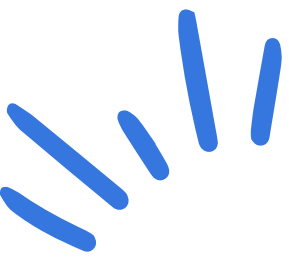
PROJECT 1: Efficient Attention Impact on Transformer Backbones for Perceptual Similarity



Università degli Studi di Roma “La Sapienza”

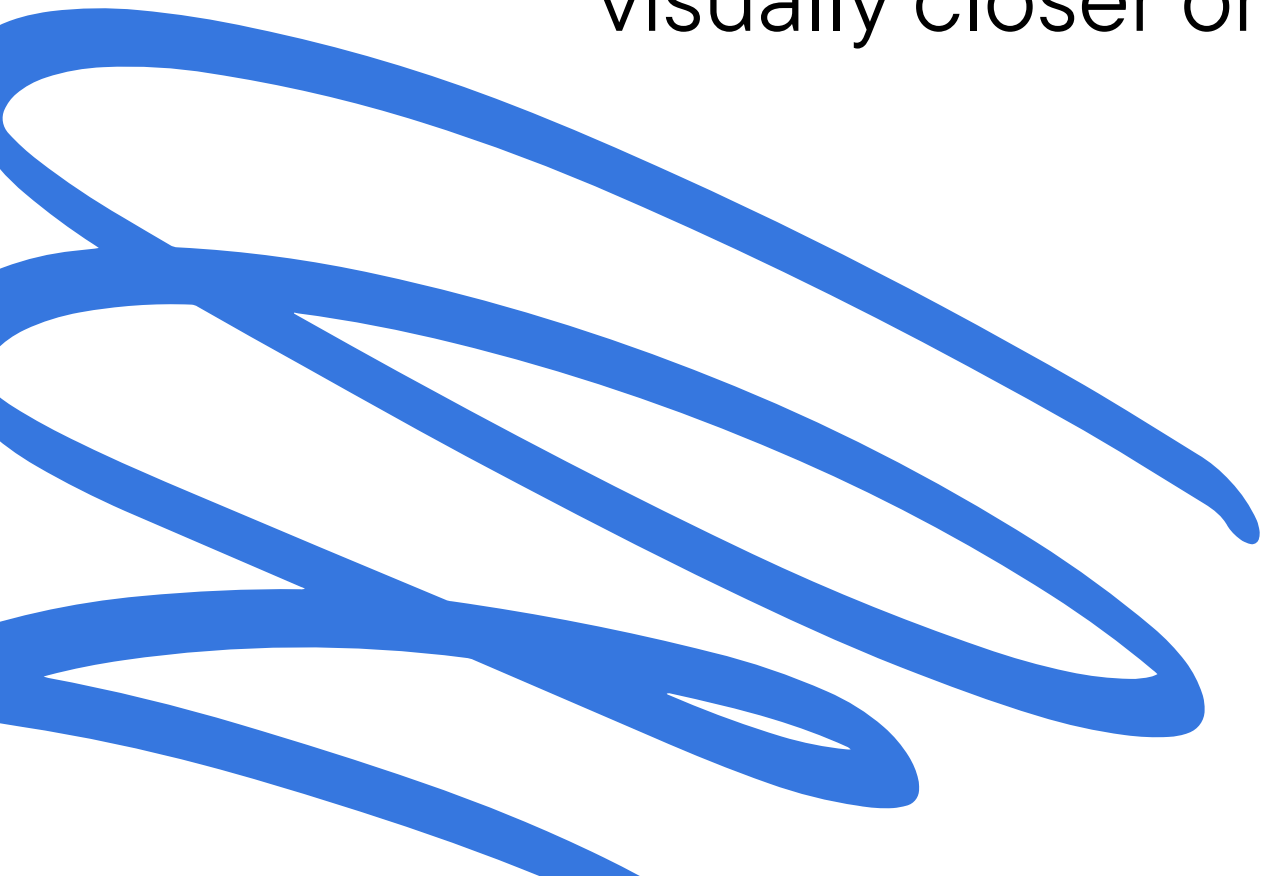
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Project Objective: The primary goal of this project is to develop a deep learning architecture capable of solving a Two-Alternative Forced Choice (2AFC) task using the NIGHTS dataset.

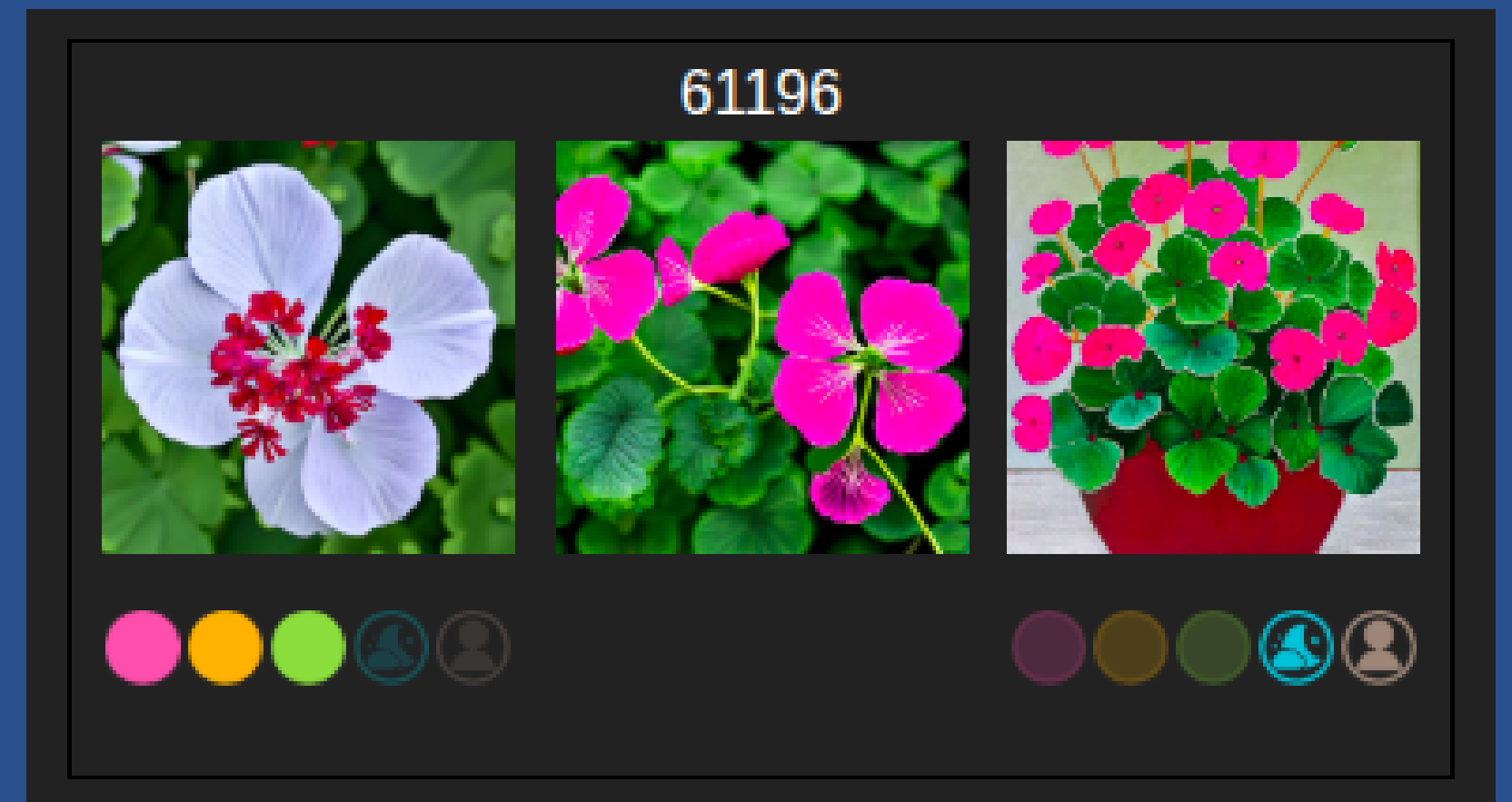
The Challenge: Our model is designed to replicate human perceptual judgments. Given a reference, the network must accurately predict which of two candidate images humans consider visually closer or more similar.





The **NIGHTS** dataset collects human judgments of visual similarity. To train the model, the data is organized into triplets consisting of a Reference image, a Left image, and a Right image.

The core task is to solve a forced choice: determining which of the two candidate images aligns closer with the human perception of visual similarity relative to the reference.



Preprocessing and Data Integrity



The dataset was divided into an 80/10/10 split for training, validation, and testing.

Data is loaded and processed efficiently using 50 data chunks to handle the large-scale triplet structures.

Quality filter: triplets kept only if voted by ≥ 6 people, with unanimous left/right consensus.

Transformations, such as horizontal flipping, are applied identically across all three images within a given triplet. This synchronized approach ensures that spatial geometry is preserved, preventing any artificial bias from disrupting the relative visual similarity.



Architecture & Models



To establish a robust baseline, we implemented a pre-trained Vision Transformer, specifically the [**vit_base_patch16_224.dino**](#) architecture.

This backbone serves as a feature extractor, generating dense embeddings for the reference image and both candidates so the model can compute their cosine distance.

To preserve the pre-trained knowledge and prevent overfitting, we applied a targeted fine-tuning strategy by unfreezing only the final two blocks of the network.



Model 0: Baseline

- Fully frozen DINO ViT – no training at all
- Measures how well pretrained features already align with human perception
- Validation accuracy: 90.8%
- Test accuracy: 88.35%

Model 1: Standard Attention

- Unfreeze last 2 transformer blocks, reimplement attention as a trainable drop-in module
- Only 16.5% of parameters trainable (14.2M / 85.8M)
- 6 epochs, AdamW (cosine LR schedule)
- Validation accuracy: 91.2%

Model 2: Mixture-of-Heads (MoH) Attention

- Replaces standard attention in the last 2 blocks with sparse, routed attention
- 4 heads total: 2 shared (always active) + 2 routed (top-k selected per token via learned gate)
- Auxiliary load-balancing loss added to the main loss
- Validation accuracy: 92.0%

Efficient Attention



Mixture-of-Heads (MoH)

- Standard attention uses all heads uniformly for every token
- MoH replaces attention in the last 2 (unfrozen) blocks only
- Heads split into: Shared (always active, global context) + Routed (dynamically selected per token)
- Auxiliary load-balancing loss (weight = 0.01) to prevent routing collapse



Results & Analysis

Our evaluation was conducted in a GPU-limited environment using Google Colab to test the model's efficiency under constrained resources.

The primary metric is 2AFC Accuracy, which measures how closely the model's predictions align with human visual similarity judgments.

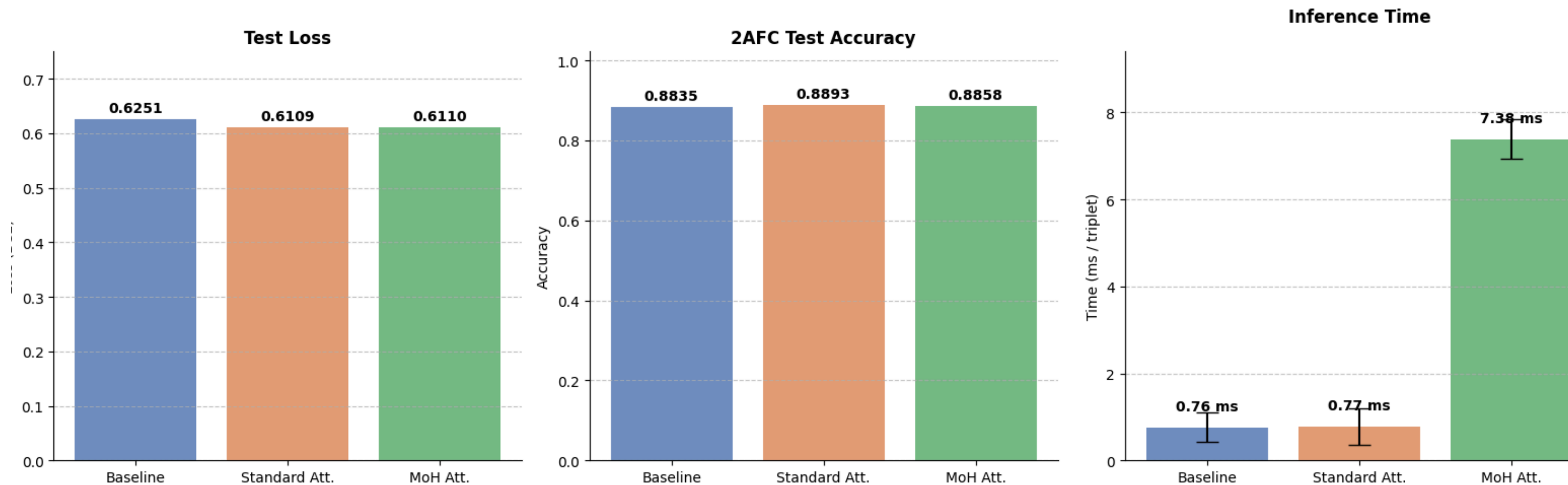
Alongside accuracy, we tracked inference speed to directly quantify the computational cost of our efficient attention variant compared to the standard baseline.



RESULTS AND ANALYSIS

All three models achieve comparable test loss and accuracy (~88-89%), with Standard Attention slightly ahead.

MoH matches this accuracy but at ~9-10× higher inference cost, a clear efficiency trade-off rather than a gain.

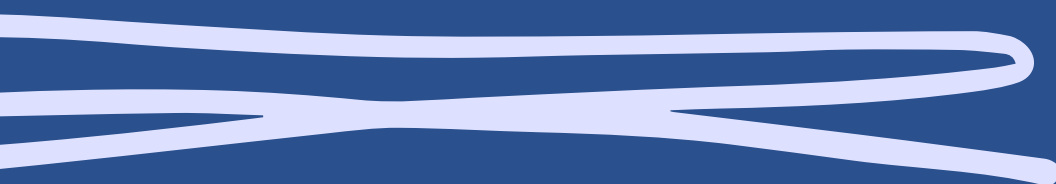


Conclusion

In conclusion, we integrated a Mixture-of-Heads attention mechanism into a perceptual similarity pipeline based on a pretrained ViT backbone.

Fine-tuning only the last two transformer blocks already improves alignment with human judgments over the zero-shot baseline, and MoH reaches the highest validation accuracy among the tested configurations.

However, this comes with a clear efficiency trade-off: our MoH implementation is significantly slower at inference than standard attention, showing that accuracy gains do not automatically translate into computational benefits, an important takeaway for future work on optimized routing implementations.



LIMITATIONS & FUTURE WORK

- Full ablation not performed due to compute constraints
 - Would test: full-network vs. partial fine-tuning
 - Would test: symmetric vs. asymmetric head split
 - Would test: impact of removing the auxiliary loss (routing collapse check)
 - MoH's inference slowdown suggests routing overhead in unoptimized implementation worth profiling/optimizing in future work
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Thank you for the attention!

